

■■ CCNA Enterprise Network & Cloud Security Drift Detector

Final Year Project Master Architecture & Feature Documentation

1. Executive Overview (Final Year Project Focus)

In enterprise Computer Networks (Cisco Routers, Switches, Firewalls, Kubernetes, Cloud Gateways), engineers define network operations through **Configuration Files** (Cisco IOS running-config, JSON, YAML). Uncontrolled manual edits cause **Configuration Drift**, introducing catastrophic security vulnerabilities and network outages.

CCNA Network & Cloud Drift Detector is an AI-powered Computer Networking & Cybersecurity application designed to parse Cisco IOS router/switch configurations, connect via Netmiko SSH to physical/simulated Cisco devices, audit CCNA security protocols (ACLs, VLANs, OSPF, BGP), render Executive Before vs. After Visual Cards ('SSL = ON -> SSL = OFF'), map Network Blast Radius, compute Risk Scores (0-100), predict future attack patterns, and execute 1-click Netmiko, Cisco CLI & Ansible auto-remediation commands.

2. Complete CCNA & Cybersecurity Feature Matrix

Feature Category	Included Capabilities & Description
■ Netmiko SSH Collector ■	Fetches live Cisco running-config via Netmiko SSH ('ConnectHandler(device_type=cisco_ios)').
■ Netmiko SSH Self-Heal ■	Autonomous self-healing execution via Netmiko SSH ('send_config_set()') to update Cisco NVRAM memory.
■ Executive Visual Diff ■	Renders intuitive Before vs. After visual badges ('SSL = ON ■ -> SSL = OFF ■') for non-technical managers.
■ CCNA Topology Blast Radius ■	Visualizes router/switch node reachability & generates Wireshark packet capture filters ('tcp.port == 9090').
■ Predictive Risk Vision ■	Predicts future multi-stage attack patterns (e.g. Port Edit -> Firewall Bypass -> Backdoor Account Creation).
■ GitOps CI/CD Shield ■	Blocks deployment pipelines if Risk Score > 50 & creates Auto-Healing Pull Requests on GitHub.
■■ Network Protocol Audit	Automated threat detection for ACL bypass ('permit ip any any'), VLAN hopping, & OSPF/BGP routing loops.
■ 'Ask Drift AI' Assistant	Typo-tolerant conversational AI assistant providing real-time Linux CLI & Cisco SRE fix advice.

3. Differentiators vs. Traditional Tools

Dimension	Standard Tools (Git Diff, Meld)	CCNA Network Drift Detector
Netmiko SSH Automation	No (Manual file copy only)	Netmiko SSH Live Router Fetching & Netmiko Auto-Heal
Executive Visual State View	No (Raw code diff lines only)	Before vs. After Visual Badges (SSL = ON -> OFF ■)
Network Packet Forensics	No	Wireshark Packet Capture Filters & Blast Heatmap
GitOps CI/CD Integration	No (Manual diffs only)	Automated Pipeline Quality Gate & Auto-PR Bot
Predictive Threat Analytics	No (Current state only)	Multi-Stage Attack Sequence Prediction
Auto-Fix Script	No	Netmiko SSH, Cisco IOS CLI, Bash & Ansible